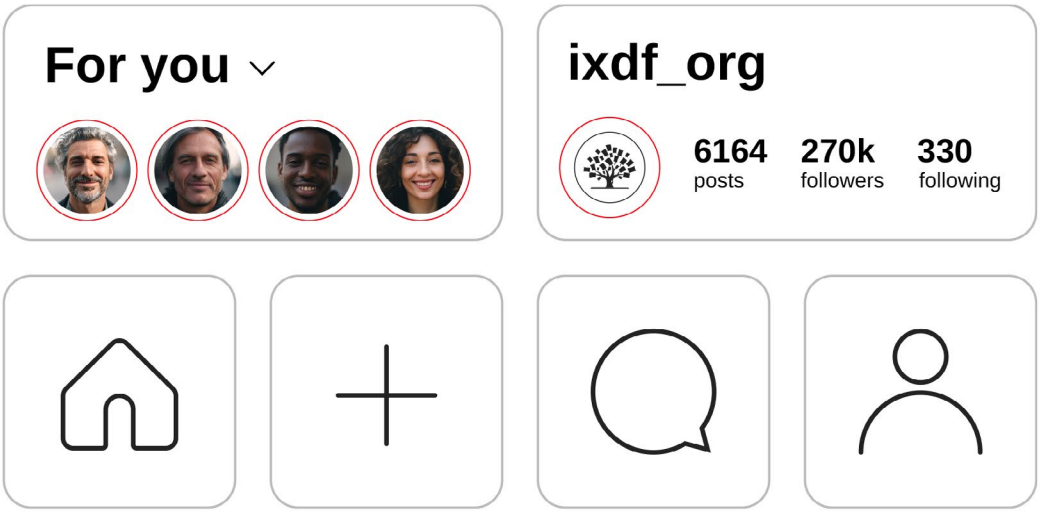


Instagram as an Object-Oriented UI

What Your Favorite Apps Can Teach You About Designing with Objects

When you're first discovering object-oriented UI (user interface) design, it can feel abstract, like trying to conduct an orchestra before you know what each instrument does. That's why this guide uses something familiar: Instagram.

By looking at object-oriented UI design through Instagram's features and interactions, you'll see how object-oriented thinking shows up in the apps we use every day.

How To Use This Template

Use this reference as your friendly guide whenever you need a quick refresher on key object-oriented UI design concepts. Come back to it when:

- You're learning a new object-oriented UI design concept and want to see how it applies in real life.
- You're exploring how object thinking can improve your design structure and usability.
- You're building or refining a UI system and want to keep things consistent and logical.

Keep it close. It'll help you connect the dots between thoughtful design and seamless experiences faster than you can double-tap a post.

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

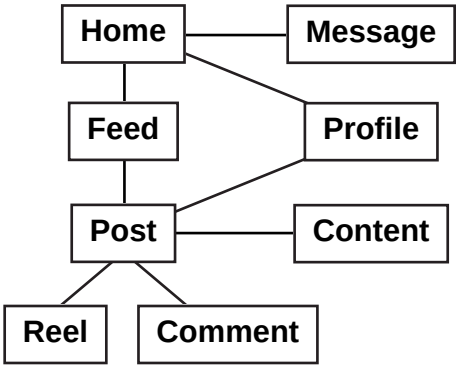
What is Object-Oriented User Interface Design?

Object-oriented user interface design is an approach to designing digital products that match how people naturally think: in terms of objects and what they can do. An object might be something concrete, like a “hotel room” or a “shopping basket,” or something more abstract, like an “account” or a “reservation.”

In object-oriented UI design, you identify these important objects in the problem space and then build the user interface around them. Instead of presenting users with a jumble of features, you give them a clear structure that **mirrors their own mental model**. That way, when someone uses your interface, it feels logical and easy to understand.

An Every Day Object-Oriented UI Example: Instagram

Although it may not be immediately apparent, many apps and websites that offer a good user experience have been designed using object-oriented (OO) programming concepts. The designers may not even be aware that they’ve done this. Still, when you utilize a user’s conceptual model from the start, the result is a clear and coherent user interface that speaks for itself. Instagram is a prime example of this. The users’ conceptual model is the same for the website and the app. Below is a basic model of the main components.

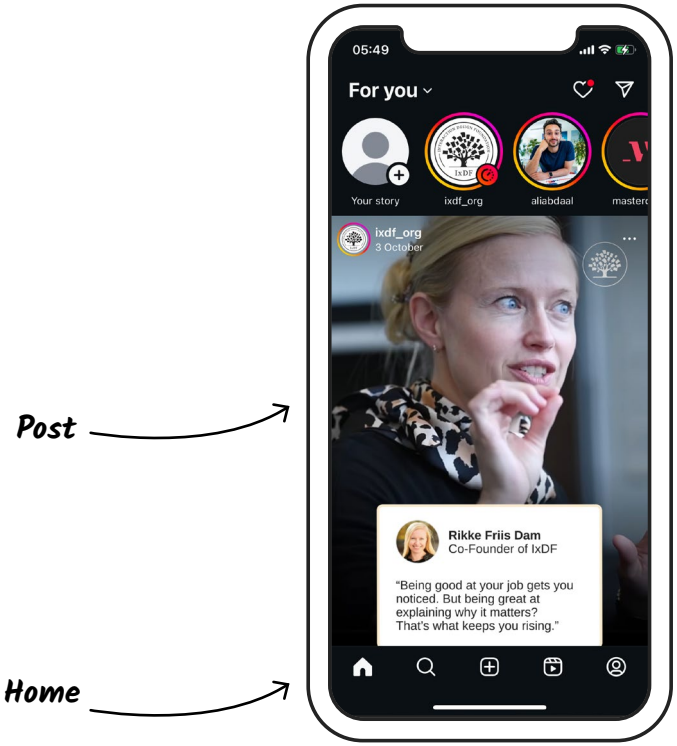


This simple model has widespread use in many fields and consequently has various names, including block diagram, entity-relationship diagram, conceptual model, and object map/model.

The concepts are the objects that Instagram users are aware of and interact with through the interface. The connecting lines represent unspecified relationships between them. This is the correct approach when you start to address a problem through an OO lens. You want to first focus on the ideas and terminology that you think users will be familiar with or readily understand.

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

OO Relationships in Instagram

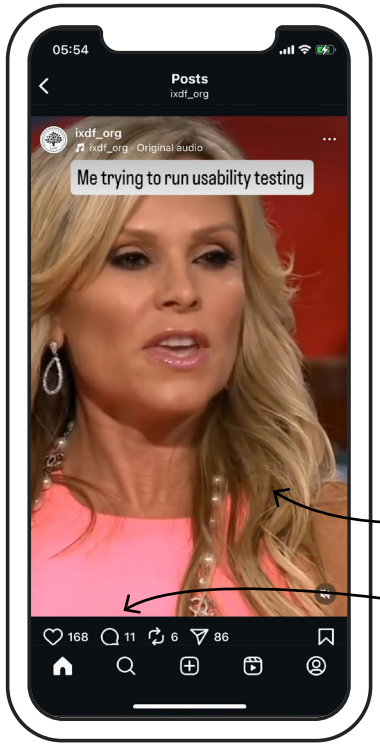


Home is where users start unless they've followed a link to a specific post or profile. The usual view is of the user's Feed, so you can think of Home as **having** or **containing** the feed, as shown in the image here. In this case, the feed starts with a post from IxDF. You'll notice that Home contains the main navigation panel with more objects than the diagram above. However, for simplicity's sake, the focus here is on the main structure.

The **feed** isn't explicitly labeled in Instagram, but it's a term commonly used and referred to in Instagram's help system. The feed **contains** a variety of **posts**. Some of those posts are reels (short video clips), so **reels** are a kind of post.

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**.
If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

OO Relationships in Instagram



Posts consist of content. This is easy to say, but it means that you can't have content without a post, and you cannot have a post without content. This makes content **part of** a post. The content can be text, images, or videos.

Posts can also **have** comments.

Content

Comments

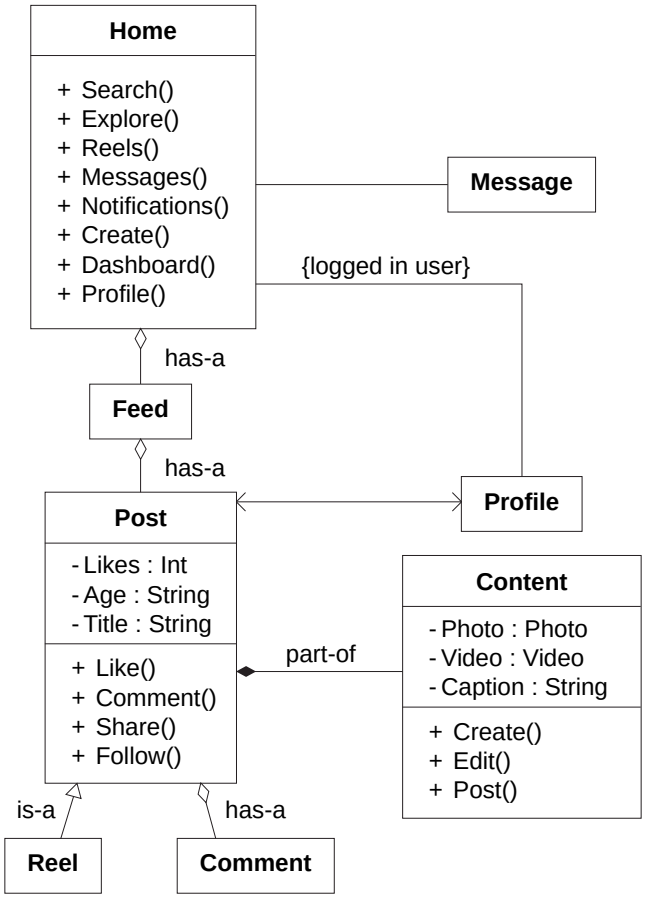
Profile

Profiles have slightly more complex relationships. A profile is **part of** every user account, and every post is **connected to** the profile of its author. So, the relationship between posts and profiles is two-way. You can access every post a user makes by visiting their profile. You can also see the profile of any user who has made a post.

Once you understand the concepts and their visual representations, much of this is easier to explain visually.

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**.
If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Instagram’s Object-Oriented UI Expanded



This more detailed diagram labels the relationships between the concepts. Where there is no label, the relationship is referred to as an association.

These are the labels used and their meanings:

Has-a: This is formally called **aggregation**, but you can also think of it as “consists of.” So, in the diagram above Home has a Feed, which has a Post. Note that in these diagrams, the concept names are always singular. If you want to, you can indicate how many feeds Home can have, and similarly, how many Posts a Feed can have. For conceptual modelling, it is common to leave these “multiplicities” out.

Part-of: This is a stronger relationship than aggregation, called **composition**. It means you cannot have the whole without its part(s). As mentioned above, you cannot have a Post without content, and you cannot have Content without a Post.

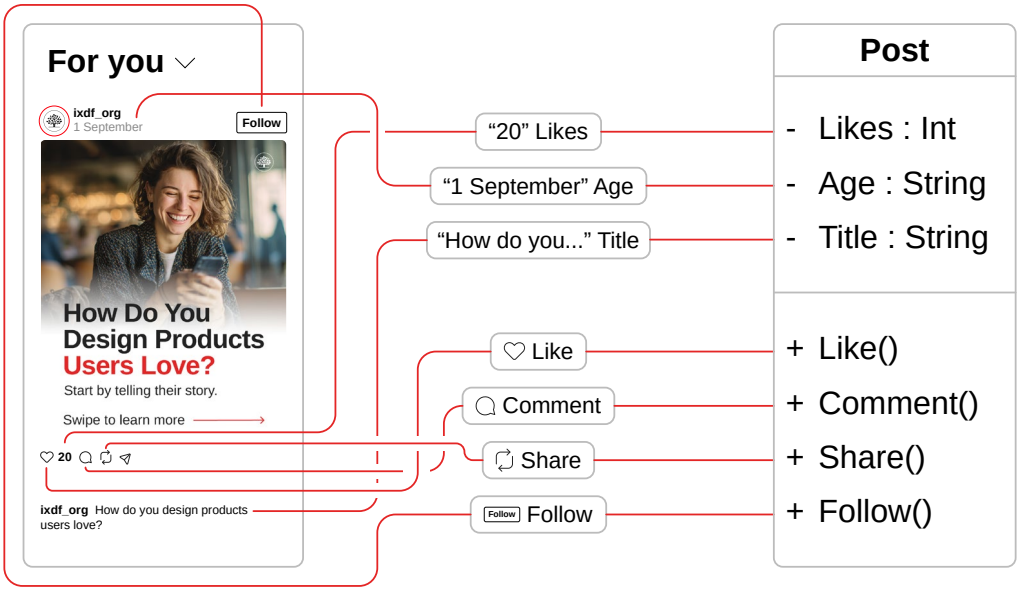
Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Is-a: As mentioned above, a Reel is a type of Post. That is the meaning of ‘is-a’, formally known as **inheritance**. You could have various Post types, but Instagram has stuck with one (so far). Everything that applies to Post also applies to a Reel, **not** vice versa. In this case, Reels are a type of video post with restrictions on format and length.

{constraint}: There is only one constraint on our relationships, which is “logged-in user,” shown in the top right. This is intended to explain that if you click on “Profile” in the left-hand navigation, you will be shown the current user’s profile.

Objects Have Attributes and Operations

The diagram also includes additional details in the form of attributes and operations, which are illustrated here for Post objects.



Attributes are characteristics of an object that you expect users to **see** or **edit**. They do this through **Views**, which are the interface windows, pop-ups, or dialogs that allow interaction. To keep the diagram simple, views are not included in high-level models.

As shown in the Post object above, when attributes are present, they always follow the object name. If you choose to display attributes in your modeling tool and there are none, an empty box will appear under the object name. (See the Home object in the more complete diagram, above.)

Operations are things that **users can do with an object**. Again, you need to create a View to allow interaction, but in this case, there will likely be only one view to display a Post in detail, including its attributes and the operations that can be performed. Operations are always displayed under attributes and always include parentheses. This is in keeping with the notation used in most programming languages, where functions or methods have parentheses.

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Learn More. Change Your Life

Learn how to use this template to its full potential in our online course **Object-Oriented UI Design: Build Interfaces Users Love.** Sign up for it today and learn how to create effective personas from scratch if you're not already enrolled.

Intermediate Course

In this course, you will:

- **Get excited** to transform messy requirements into smooth, user-centered solutions people instantly love. **Most designers have never even heard of object-oriented UI design, the new standard in UX.** This gives you an edge. Research shows that clear, intuitive UI **can double conversion rates**. Object-oriented UI design makes that easy.
- **Make yourself invaluable** by keeping design and development in sync. You'll discover how to avoid miscommunication, impractical proposals, and expensive redesigns, and get **recognition** as the person everyone can **trust to drive real impact with critical projects.**
- **Gain confidence and credibility** as the designer who cuts through complexity and delivers clarity. You'll master how to analyze needs, prevent feature creep, and create interfaces that **stay consistent, intuitive, and scalable.**
- **Craft your personal portfolio** with results that prove your value. You'll create career assets like conceptual models and design maps that show how you can move from requirements to **implementation-ready UIs that users love.**

Learn more about this course >

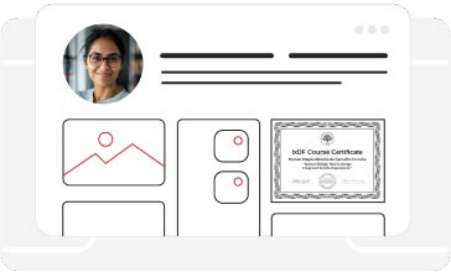
Design a Life You Love

IxDF — Global Leader in Design Education. Founded in 2002.

ixdf.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.

Make Yourself Valuable.
In Any Job. Any Industry



Get a Course Certificate

Build your credibility with IxDF Certificates on your LinkedIn profile and resumé. Be recognized as part of an elite group and prove that you've learned from industry leaders.

Learn From the Best

Learn design from experts at NASA, Google and Adobe. Get extra help from our global community and insider knowledge from industry leaders. Use it to create your own career success.

Build a Standout Portfolio

Build your portfolio to showcase your skills. Our courses make it easy for you to build an impressive portfolio with templates and step-by-step guides.

See All IxDF Courses ›

Global Leader in Design Education

Turn your passion into your paycheck. In any job. Any industry.

Design isn't just for designers. It's how you create products people love, delight users, win customers, and drive growth. More love. More loyalty. More profits.

Over 1.2 million enrollments since 2002. Trusted by designers worldwide to build careers, create impact, and shape the future of design.

Design a Life You Love

IxDF — Global Leader in Design Education. Founded in 2002.

ixdf.org

Creative Commons BY-SA license: You are free to edit and redistribute this template, **as long as you give credit to IxDF**. If you remix, transform, or build upon this template, you must distribute it under the same CC BY-SA license.